20 An estimate of the height, H metres, of a tall building can be found using the formula

$$H = 4f + 12$$

where the building is f floors high.

A tall building is 110 floors high.
The real height of the building is 442 m.

Seb uses the formula to find an estimate of the height of this building.
He then finds the difference between his estimate and the real height.

Show that this difference is less than 5% of the real height.

Estimate = 4 × 110 + 12 = 452 m

Difference = 452 – 442 = 10 m

5% of real height = 0.05 × 442 = 22.1 m

10 m < 22.1 m, so the difference is <5%.

4 marks total
Estimate 452 m
Difference 10 m
5% is 22.1 m
Show 10 < 22.1

(Total for Question 20 is 4 marks)